# Sang Hyuk Hong

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## EDUCATION

**Korea University** **March 2019 – February 2023**
*B.S. in Physics* *Seoul, Republic of Korea*
GPA: 4.09 / 4.50 (Major GPA: 4.19 / 4.50)

## RESEARCH EXPERIENCE

**Seoul National University** **July 2026 – Present**
*Research Assistant | PI: Prof. Yong-il Shin* *Seoul, Republic of Korea*

- Constructed coil and magnet systems for 2D and 3D magneto-optical traps (MOTs) of a dual-species Rb–Yb tweezer array platform.
- Built the high-vacuum system for an ultra-low-expansion (ULE) reference cavity to stabilize the 399, 556, and 780 nm cooling lasers of a quantum-degenerate Rb–Yb mixture experiment.
- Implementing electronic sideband offset Pound–Drever–Hall (PDH) locking of the 1112 nm laser to the ULE cavity for isotope-selective narrow-line cooling of fermionic $^{173}$Yb and bosonic $^{174}$Yb.

**Agency for Defense Development** **June 2023 – May 2026**
*Research Officer for National Defense | Lieutenant* *Daejeon, Republic of Korea*

- Realized navigation-grade low-SWaP (size, weight, and power) nuclear magnetic resonance gyroscopes (NMRGs) with bias stability under 0.02 deg/h [5], enabling in-house inertial measurement unit (IMU) and inertial navigation system (INS) development.
- Demonstrated laser mode-hopping assisted single-beam all-optical scalar optically pumped magnetometers (OPMs) [4] for applications in vector magnetometry, magnetic navigation, magnetic anomaly detection, and magnetic field imaging.
- Established novel fabrication techniques for atomic gas cells, enabling atomic sensors with compact form factors [3], prolonged cell lifetimes [2], and improved geometric symmetry and pressure reproducibility [1].
- Formulated theoretical and numerical frameworks for spin relaxation and frequency shifts induced by time-periodic magnetic-field inhomogeneities [6].

## ACADEMIC DISTINCTIONS

**National Scholarship for Study Overseas** **2027 – 2029**
*National Institute for International Education*

- Secured funding of up to US$40,000/year for the first two years of graduate study.

**KFAS Study Abroad Program** **2026**
*Korea Foundation for Advanced Studies (KFAS)*

- Selected as one of 45 participants nationwide.

**Commendation from the Minister of DAPA** **2026**
*Defense Acquisition Program Administration (DAPA)*

- Awarded for outstanding research achievements during military service.

**National Scholarship for Natural Science and Engineering** **2021 – 2022**
*Korea Student Aid Foundation*

- Full tuition coverage plus stipends totaling US$20,000.

## PUBLICATIONS

### Journal Articles

[1] **S. H. Hong**, Y. Lim, S. H. Yim, S. Lee, H. Kim, T. Jeong, and J. B. Nam, "Wire-sealed atomic gas cells for noble-gas nuclear magnetic resonance applications," Rev. Sci. Instrum. **97**, 103203 (2026).

[2] H. Kim, T. Jeong, **S. H. Hong**, J. B. Nam, S. Lee, Y. Lim, and S. H. Yim, "Aging test of an atomic vapor cell with $Al_2O_3$ wall coating on cubic glass," Appl. Opt. **64**, 7932–7937 (2025).

[3] **S. H. Hong**, S. H. Yim, T. Jeong, J. B. Nam, and S. Lee, "Fabrication of an Rb–Xe gas cell with optical components attached by optical contact bonding for an atom spin gyroscope," Appl. Opt. **64**, 3988–3994 (2025).

### Preprints

[4] J. H. Yoon, **S. H. Hong**, T. Jeong, S. H. Yim, K. M. Shim, and S. Lee, "Laser mode-hopping assisted all-optical single beam pulsed atomic magnetometer," submitted to Phys. Rev. A, arXiv:2404.01874v2 (2025).

### Manuscripts in Preparation

[5] **S. H. Hong**, S. Lee, S. H. Yim, S. R. Yeo, S. Lee, H. Yu, T. Jeong, J. B. Nam, D. Kim, and Y. Lim, "Development and characterization of atom spin gyroscopes," manuscript in preparation.

[6] **S. H. Hong** and S. Lee, "Spin relaxation and frequency shift from time-periodic magnetic-field inhomogeneities in diffusing spin-1/2 gases," manuscript in preparation.

## PATENTS

- **Heated-wire sealing method for atomic vapor cells**
  KR Patent pending, 2025.
- **Vector magnetic-field measurement method using AC Stark effect in a scalar optically pumped magnetometer**
  KR Patent pending, 2026.

## SELECTED PRESENTATIONS

**2025 Korean Physical Society Fall Meeting** **October 2025**
*Oral Presentation* *Gwangju, Republic of Korea*

- AC magnetic-field-gradient-induced transverse relaxation for diffusive spin-1/2 particles in a cubic cell

**16th CLEO-PR** **August 2024**
*Poster Presentation* *Incheon, Republic of Korea*

- Atom spin gyroscope using an Rb–Xe gas cell

**55th APS DAMOP Meeting** **June 2024**
*Poster Presentation* *Fort Worth, Texas, USA*

- Analysis of heading errors in pulsed-mode and continuous-mode all-optical atomic magnetometers

**2023 Korean Physical Society Fall Meeting** **October 2023**
*Poster Presentation* *Changwon, Republic of Korea*

- Optimization of saddle coil for uniform magnetic field in atomic magnetometer experiment

## TECHNICAL SKILLS

**Scientific Computing** Python, Wolfram Mathematica.

**Scientific Software** AutoCAD, COMSOL Multiphysics, LabVIEW, Origin, SolidWorks.